



OSI Model (Open Systems Interconnection)

The OSI model is a reference framework created by ISO (International Standards Organization) to understand and design network communication.

It breaks the communication process into **7 layers**, each with its own role.

1. Physical Layer (L1)

 What it does:

- Sends **bits (0s and 1s)** as signals (electric, light, or radio).
- Just the **physical connection**.

 Example:

- Wi-Fi signal, Mobile 4G signal, Cable carrying internet.
- Like the **road** where cars travel.

2. Data Link Layer (L2)

 What it does:

- Makes sure data moves **from one device to another** in the same network.
- Uses **MAC address** (unique ID of each device's network card).



Example:

- Your laptop sends data to your Wi-Fi router using MAC address.
- Like writing **house number** on a letter in the same street.

3. Network Layer (L3)



What it does:

- Decides the **path** to send data across networks.
- Uses **IP address** (location of device on the internet).



Example:

- Sending an email to USA → Routers use IP addresses to find the correct server.
- Like the **post office** finding the correct city and country.

4. Transport Layer (L4)



What it does:

- Makes sure data reaches the right **application (port)**.
- **TCP**: Reliable (no missing parts).
- **UDP**: Faster, but some parts may miss.



Example:

- **TCP**: Online banking → all details must arrive safely.
- **UDP**: PUBG game → speed is important, small loss is ok.
- Like choosing **courier service** (safe delivery vs fast delivery).

5. Session Layer (L5)



What it does:

- Opens and manages the **session (connection)** between two devices.

- Keeps the conversation alive until done.



Example:

- WhatsApp Web → once you scan QR, a session is created and stays open.
- Like a **phone call session** — you talk until you hang up.

6. Presentation Layer (L6)



What it does:

- Prepares data so both devices understand.
- **Translation, Encryption, Compression.**



Example:

- HTTPS website → encrypts your password so hackers can't read it.
- Watching YouTube → video compressed before sending.
- Like a **translator** or **packing data in a box**.

7. Application Layer (L7)



What it does:

- The layer you **see and use**.
- Apps use this layer to talk over the network.



Example:

- Web Browser → HTTP/HTTPS.
 - Email → SMTP, IMAP, POP3.
 - WhatsApp → uses its own protocols.
 - Like the **letter itself** that you write and send.
-
- L1–L4 = How data travels.
 - L5–L7 = How apps use the data.